Evaluating a solution

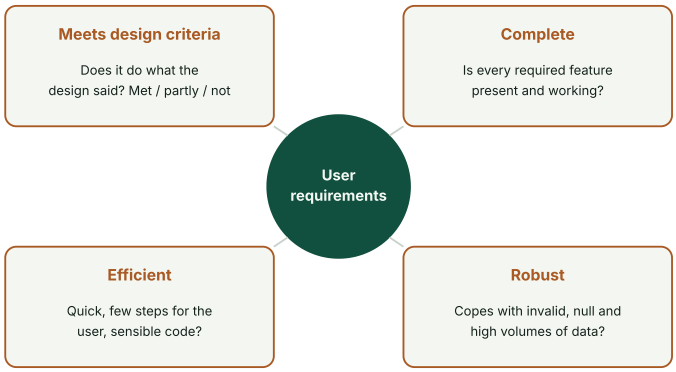
D23 FACT FILE

Specification: students should be able to

- Explain how evaluation is used to ensure that a solution: meets the original design criteria; is a full and complete solution; is an efficient solution; and is a robust solution

CCEA fact file: Testing

Key words: evaluation, user requirements, design criteria, success criteria, complete, efficient, robust, stakeholder, end user, refinement, high-volume testing



Evaluating a solution against the user requirements: meets design criteria, complete, efficient, robust

Evaluation

Evaluation measures a solution against the original user requirements and design criteria. It asks whether the solution meets the design, whether it is complete (every requirement implemented and working), whether it is efficient (fast, economical with resources and easy for the user), and whether it is robust (it copes with invalid, unexpected and large volumes of data without failing). Evidence comes from testing and user feedback. Evaluation should happen throughout development as well as at the end.

T Key terms from the mark schemes

Evaluation
Judging the solution against the user requirements and design.
Credited: U4 2019 Q11, 2024 Q9

User requirements
The agreed list of what the system must do, used to judge the solution.
Credited: U4 2025 Q10

Robustness
The ability to handle invalid, unexpected and high volumes of data without failing.
Credited: U4 2024 Q9, 2025 Q10

Stakeholders
People with an interest in the system: client, end users, developers.
Credited: U4 2019 Q11

Exam focus

2019 Q11 (purpose, stakeholders, requirements, original plan), 2022 Q11, 2023 Q11, 2024 Q9, 2025 Q10. Use the scenario and refer to requirements, testing evidence and robustness.

Unit 4 2025 Q10	[4]	User requirements; robustness
Unit 4 2024 Q9	[7]	Evaluation and robustness
Unit 4 2023 Q11	[7]	Evaluation using requirements and design documents
Unit 4 2022 Q11	[7]	Evaluation statements; complete, robust, efficient
Unit 4 2019 Q11	[10]	Evaluation: purpose, stakeholders, requirements, original plan

? Check yourself

1. Give two reasons why a solution should be evaluated. [2]
2. Explain how user requirements are used during evaluation. [2]
3. What is meant by a robust solution? [2]

Answers: 1. Any two from: to judge how well the solution solves the original problem (1) / to check it meets the user requirements/design criteria (1) / to identify weaknesses, missing features and improvements/refinements (1) / to check it is complete, efficient and robust (1) 2. The user requirements are the agreed standard the solution is measured against – each is checked as a checklist: fully met / partly met / not met (1) / if a feature is missing, the requirements show what must be added before the solution is complete (1) 3. A robust solution handles errors and unexpected input without crashing (1) / it copes with invalid, unexpected/exceptional and high volumes of data (e.g. through validation and error trapping) (1)